

EXPERIMENT NO. 7 / 8

UML Sequence Diagram & Collaboration Diagram

Project: Hospital Management System (HMS)

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1. Aim

To model dynamic time-ordered and link-oriented object interactions in the Hospital Management System (HMS) using UML Sequence and Collaboration (Communication) diagrams.

2. Objectives

- To construct a Sequence Diagram displaying chronologically ordered message exchanges across boundary, control, and entity objects.
- To construct a Collaboration Diagram highlighting structural organization and numbered message links.
- To model real-time transactions: appointment scheduling, consultation charge aggregation, and payment gateway authorization.

3. UML Sequence Diagram

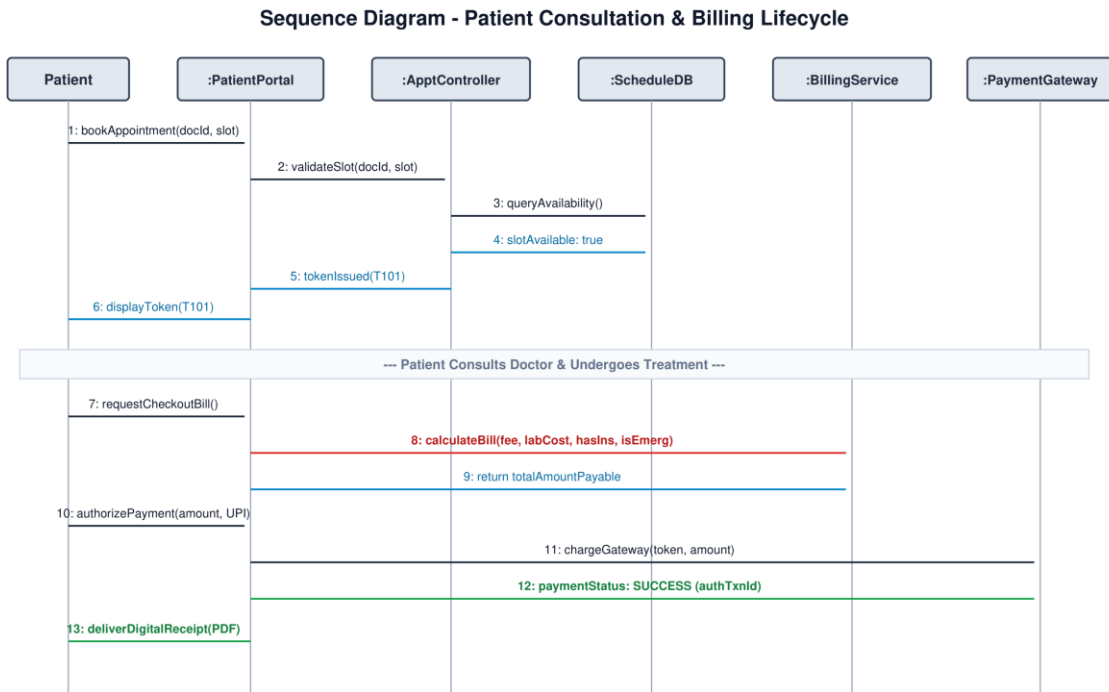


Fig 1: Sequence Diagram - Patient Consultation & Billing Lifecycle (HMS)

Message Flow Analysis:

- Messages 1–6 (Scheduling): Patient triggers bookAppointment via PatientPortal. ApptController queries ScheduleDB, reserves slot, and returns token.
- Consultation Phase: ConsultationService updates EHR records and posts consultation fee to BillingService.
- Messages 7–13 (Billing & Settlement): calculateBill executes with parameters (fee, labCost, hasIns, isEmerg). Patient authorizes UPI transaction, PaymentGateway returns approval token, and invoice is generated.

4. UML Collaboration Diagram

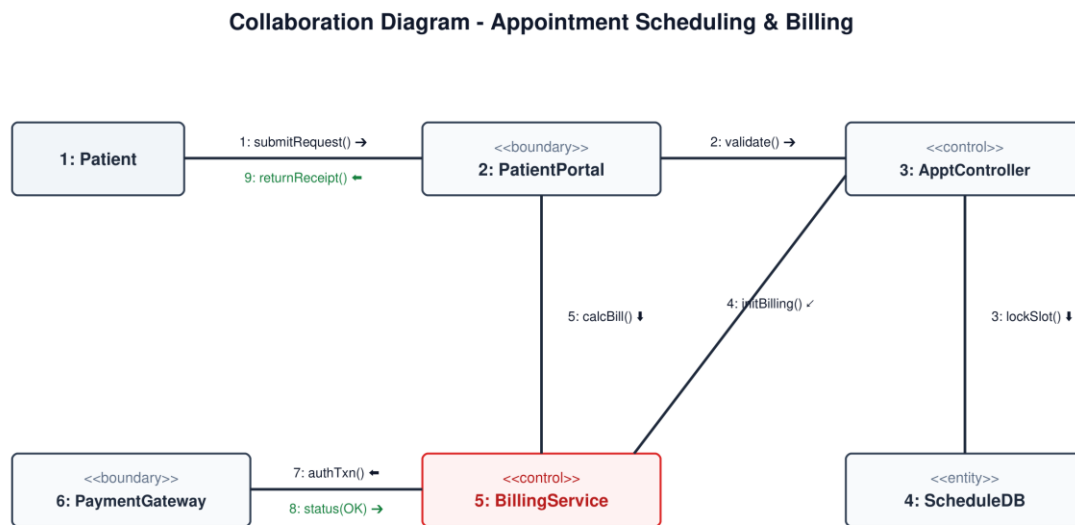


Fig 2: Collaboration Diagram - Appointment Scheduling & Billing (Link-Oriented)

Sequence vs Collaboration Diagram Comparison:

Comparison Parameter	Sequence Diagram	Collaboration / Communication Diagram
Primary Emphasis	Time sequence of messages (top to bottom timeline).	Structural relationships and links between interacting objects.
Lifelines & Activation	Explicit vertical lifeline bars showing object lifetime and active execution.	Objects represented as icons/rectangles; lifelines are omitted.
Message Numbering	Optional, as chronological order is implicit in vertical placement.	Mandatory numbered decimal prefixes (e.g., 1:, 2:, 3:) to indicate

Suitability	Best for detailed execution traces and timeout/loop modeling.	message sequence. Best for understanding object collaborations and structural impacts of design changes.
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5. Conclusion

The Sequence and Collaboration diagrams provide complementary perspectives on runtime interactions in HMS, verifying seamless coordination between scheduling, clinical recording, and financial transactions.